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Motivation

BIOMIMETIC TOP-DOWN VISION

- Human vision first forms a rough overview, then revisits salient regions for finer perception.
- Classic pyramidal backbones only move bottom-up; intermediate stages lack explicit semantic guidance.
- Pure ConvNets also struggle to combine global dependence with strong local inductive bias.

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DDS

THREE SUBNETWORKS

- Base-Net extracts shared low/mid-level features at 1/16 resolution.
- A lightweight Overview-Net immediately downsamples to form a coarse but semantic context prior.
- Focus-Net returns to the mid-level map and refines it under feedback from that global prior.

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ContMix

- Each input token measures affinity to region centers in the top-down context map.
- A linear projection converts each affinity row into spatially varying convolution weights.
- Fixed kernel geometry preserves locality, while every weight carries global semantic information.

OVERLOCK

Overview First,
Look Closely Next

Top-Down ConvNet with Context-Mixing Kernels | 2025

Give a pure ConvNet explicit top-down vision: build a cheap global overview, feed it back as context, and use it to generate dynamic local kernels for focused high-resolution perception.

top-down attentiondynamic convolutionConvNet

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Block design

GUIDED FEATURE MODULATION

- Dynamic blocks update both features and context priors; an initial-prior skip prevents semantic dilution.
- A context-guided gate modulates feature channels, and an auxiliary loss supervises Overview-Net semantics.
- Ablation raises a plain baseline from 76.3/38.8 to 80.8 Top-1/43.8 mIoU.

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Evidence

PURE CONVNET LEADS

- ImageNet: OverLoCK-T reaches 84.2% at 5.5G FLOPs/33M parameters; -B reaches 85.1%.
- COCO Cascade Mask R-CNN: OverLoCK-S reaches 53.6 APbox, +1.0 over MogaNet-B.
- ADE20K: OverLoCK-T reaches 50.3/50.8 mIoU and exceeds UniRepLKNet-T by 1.7 points.

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Meaning & limits

TAKEAWAY

- Top-down feedback can enlarge effective receptive fields and improve localization without recurrent passes.
- ContMix bridges global context and convolution's stable local bias across image resolutions.
- The three-branch design and dynamic kernels add complexity; evidence centers on standard discriminative vision tasks.